Advanced Fluid Dynamics

DAY 8

Creating the new OF solver

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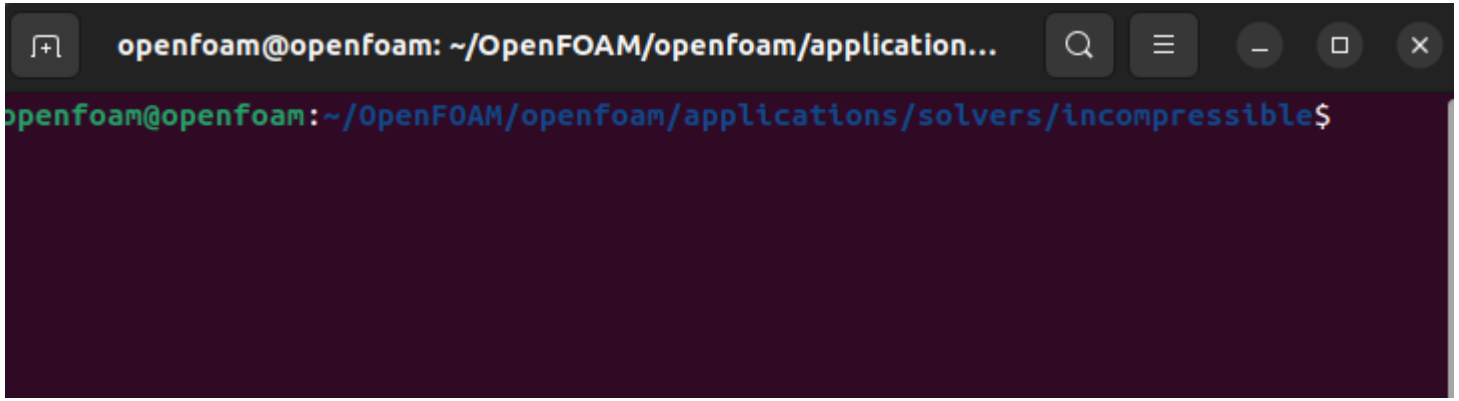
- Creating a new OpenFOAM solver based on icoFoam
- Exercise – 6

Following things to do:

1. Copy default icoFoam solver
2. Adding the temperature field in the icoFoam solver
3. Copy default cavity tutorial based on icoFoam
4. Add a new file for initial and boundary conditions
5. Add respective files in fvSchemes and fvSolution
6. Run your new case file
7. Upload your new solver, case files, and results in GITHUB

1. Copy default icoFoam solver

<u>From the location:</u>



<u>To make your own solver, firstly do the following:</u>

- Change the default icoFoam name into myIcoFoam.
- Change the source file name from icoFoam.C into myIcoFoam.C
- Open Files, and do the following:

  myIcoFoam.C

  EXE = $(FOAM_USER_APPBIN)/myIcoFoam

2. Adding the temperature field in the icoFoam solver

Under <u>createFields.H</u>

```
//Add here...
dimensionedScalar DT
(
    "DT",
    dimViscosity,
    transportProperties
);
```

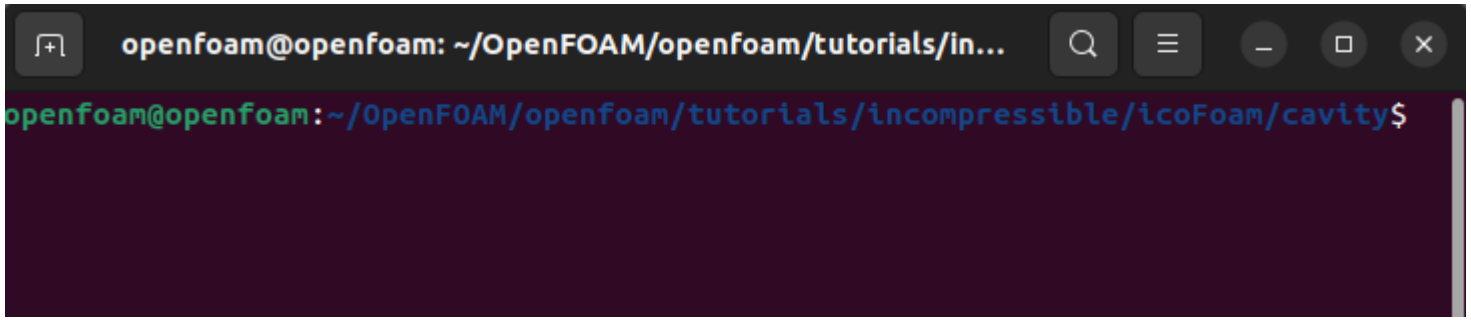
```
Info<< "Reading field T\n" <<endl;
volScalarField T
(
    IOobject
    (
        "T",
        runTime.timeName(),
        mesh,
        IOobject::MUST_READ,
        IOobject::AUTO_WRITE
    ),
    mesh
);
```

Under <u>myIcoFoam.C</u>

```
//add these lines...
        fvScalarMatrix TEqn
        (
            fvm::ddt(T)
            + fvm::div(phi, T)
            - fvm::laplacian(DT, T)
        );

        TEqn.solve();
//done adding lines...
```

3. Copy default cavity tutorial based on icoFoam

<u>From the location:</u>



Modify your solver name as myCavityCaseFile

4. Add a new file for initial and boundary conditions

Under constant/transportProperties

Add DT (new variable created)

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  v2306                                 |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      transportProperties;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

nu              0.01;

DT              0.002;

// ************************************************************************* //
```

Add “T” field in “0” file

```
/*--------------------------------*- C++ -*----------------------------------*\
| =========                 |                                                 |
| \\      /  F ield         | OpenFOAM: The Open Source CFD Toolbox           |
|  \\    /   O peration     | Version:  v2306                                 |
|   \\  /    A nd           | Website:  www.openfoam.com                      |
|    \\/     M anipulation  |                                                 |
\*---------------------------------------------------------------------------*/
FoamFile
{
    version     2.0;
    format      ascii;
    class       volScalarField;
    object      T;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

dimensions      [0 0 0 1 0 0 0];

internalField    uniform 300;

boundaryField
{
    movingWall
    {
        type        fixedValue;
        value       uniform 350;
    }

    fixedWalls
    {
        type        fixedValue;
        value       uniform 300;
    }

    frontAndBack
    {
        type        empty;
    }
}

// ************************************************************************* //
```

5. Add respective files in fvSchemes and fvSolution

Under system/fvSchemes

```
FoamFile
{
    version     2.0;
    format      ascii;
    class       dictionary;
    object      fvSchemes;
}
// * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * * //

ddtSchemes
{
    default         Euler;
}

gradSchemes
{
    default         Gauss linear;
    grad(p)         Gauss linear;
}

divSchemes
{
    default         none;
    div(phi,U)      Gauss linear;
    div(phi,T)      Gauss upwind;
}

laplacianSchemes
{
    default         Gauss linear orthogonal;
    laplacian(DT,T) Gauss linear corrected;
}

interpolationSchemes
{
    default         linear;
}

snGradSchemes
{
    default         orthogonal;
}


// ************************************************************************* //
```

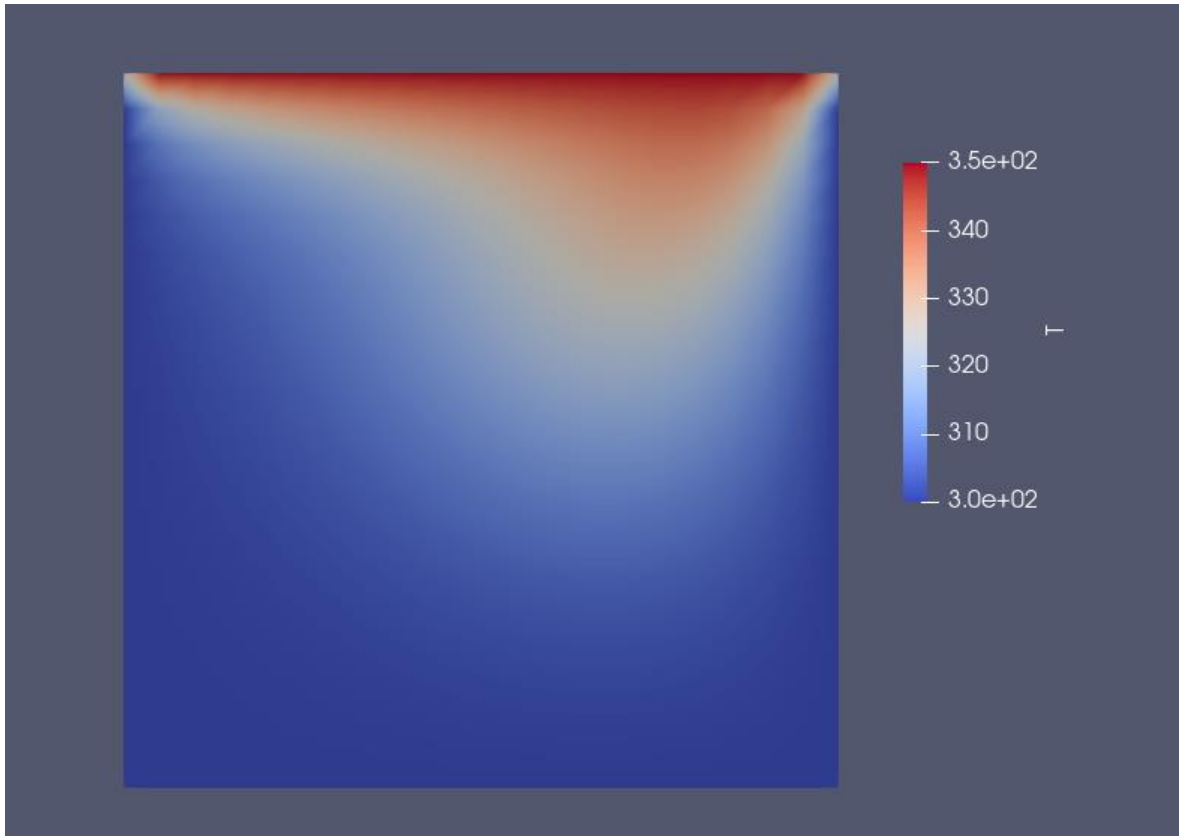
Under system/fvSolution

```
solvers
{
    p
    {
        solver          PCG;
        preconditioner  DIC;
        tolerance       1e-06;
        relTol          0.05;
    }

    pFinal
    {
        $p;
        relTol          0;
    }

    T
    {
        solver          PBiCGStab;
        preconditioner  DILU;
        tolerance       1e-6;
        relTol          0.1;
    };

    U
    {
        solver          smoothSolver;
        smoother        symGaussSeidel;
        tolerance       1e-05;
        relTol          0;
    }
}

PISO
{
    nCorrectors     2;
    nNonOrthogonalCorrectors 0;
    pRefCell        0;
    pRefValue       0;
}

// ************************************************************************
```

6. Run your new case file

```
Time = 0.4998

Courant Number mean: 0.00444052 max: 0.0170421
smoothSolver:  Solving for Ux, Initial residual = 1.5517e-09, Final residual = 1.5517e-09, No Iterations 0
smoothSolver:  Solving for Uy, Initial residual = 2.04474e-09, Final residual = 2.04474e-09, No Iterations 0
DICPCG:  Solving for p, Initial residual = 1.12694e-06, Final residual = 1.80179e-08, No Iterations 1
time step continuity errors : sum local = 6.5149e-13, global = -9.77893e-21, cumulative = -2.73277e-19
DICPCG:  Solving for p, Initial residual = 1.57304e-08, Final residual = 1.57304e-08, No Iterations 0
time step continuity errors : sum local = 5.69014e-13, global = -1.06475e-20, cumulative = -2.83925e-19
DILUPBiCGStab:  Solving for T, Initial residual = 2.59838e-05, Final residual = 5.04678e-10, No Iterations 1
ExecutionTime = 2.86 s  ClockTime = 3 s

Time = 0.4999

Courant Number mean: 0.00444052 max: 0.0170421
smoothSolver:  Solving for Ux, Initial residual = 1.56123e-09, Final residual = 1.56123e-09, No Iterations 0
smoothSolver:  Solving for Uy, Initial residual = 2.05631e-09, Final residual = 2.05631e-09, No Iterations 0
DICPCG:  Solving for p, Initial residual = 5.45261e-07, Final residual = 5.45261e-07, No Iterations 0
time step continuity errors : sum local = 1.96933e-11, global = 9.276e-21, cumulative = -2.74649e-19
DICPCG:  Solving for p, Initial residual = 5.61721e-07, Final residual = 5.61721e-07, No Iterations 0
time step continuity errors : sum local = 2.02877e-11, global = 2.35267e-20, cumulative = -2.51122e-19
DILUPBiCGStab:  Solving for T, Initial residual = 2.597e-05, Final residual = 5.04411e-10, No Iterations 1
ExecutionTime = 2.86 s  ClockTime = 3 s

Time = 0.5

Courant Number mean: 0.00444052 max: 0.0170421
smoothSolver:  Solving for Ux, Initial residual = 1.41142e-09, Final residual = 1.41142e-09, No Iterations 0
smoothSolver:  Solving for Uy, Initial residual = 1.86002e-09, Final residual = 1.86002e-09, No Iterations 0
DICPCG:  Solving for p, Initial residual = 1.02705e-06, Final residual = 1.63329e-08, No Iterations 1
time step continuity errors : sum local = 5.92229e-13, global = -6.97975e-21, cumulative = -2.58102e-19
DICPCG:  Solving for p, Initial residual = 1.42803e-08, Final residual = 1.42803e-08, No Iterations 0
time step continuity errors : sum local = 5.18098e-13, global = 2.40213e-20, cumulative = -2.34081e-19
DILUPBiCGStab:  Solving for T, Initial residual = 2.59562e-05, Final residual = 5.04142e-10, No Iterations 1
ExecutionTime = 2.86 s  ClockTime = 3 s

End

openfoam@openfoam:~/Documents/myCavityCaseFile$
```

7. Upload your new solver, case files, and results in GITHUB



Exercise – 6

[Exercise-6] Creating a new OpenFOAM solver based on icoFOAM by adding a new variable (Temperature) #10

Kumaresh1176 started this conversation in General

Kumaresh1176 5 minutes ago Maintainer

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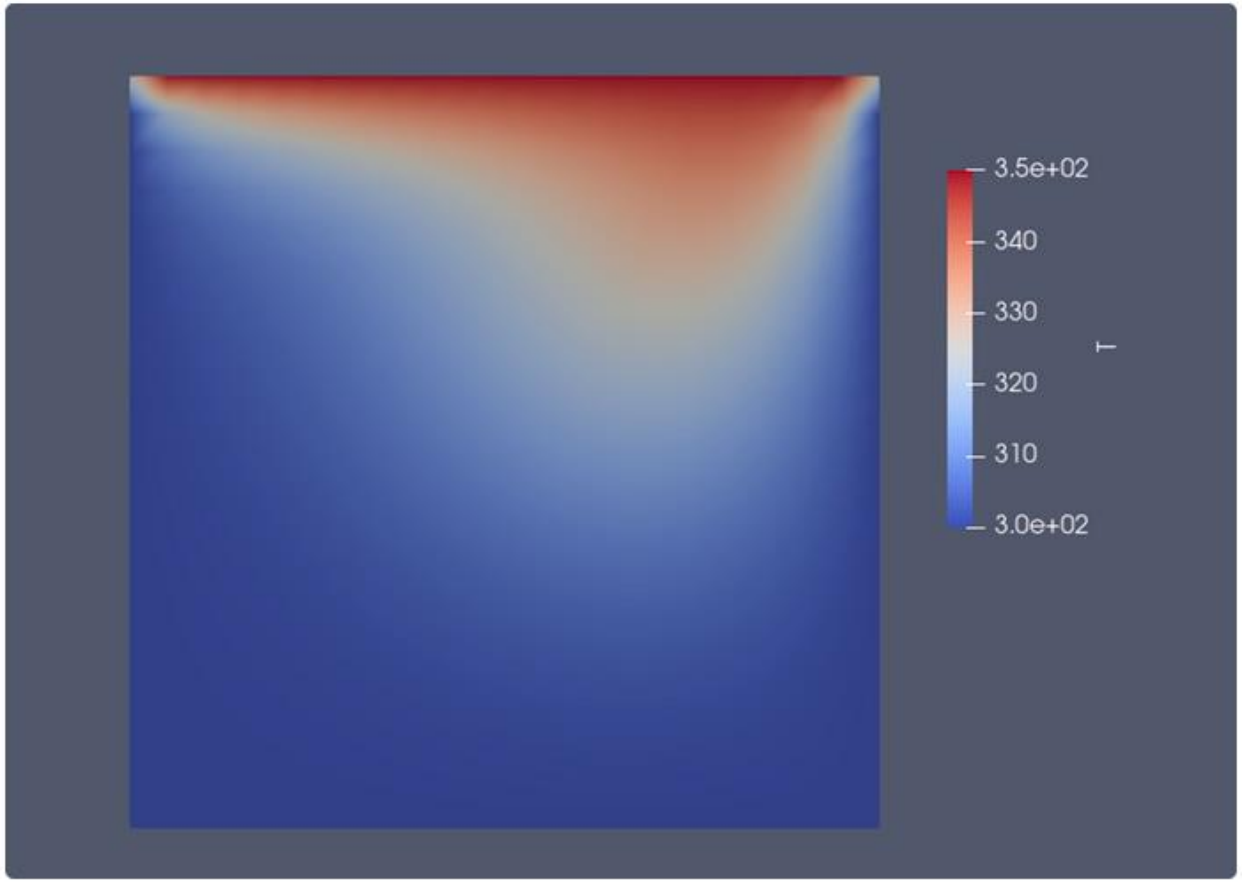

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